Chapter 3: Conditional Statements

Python Programming Study Guide Series

CONTROL FLOW BASICS

Conditional statements enable a program to execute different logic paths depending on whether specified conditions evaluate to `True` or `False`.

```
# Decision Tree Structure
temperature = 28
if temperature > 30:
    print("It is a hot day.")
elif temperature >= 20:
    print("It is a pleasant day.")
else:
    print("It is a cold day.")
```

OPERATORS IN CONDITIONALS

Category	Operators	Example Expression	Result
Comparison	==, !=, >, <, >=, <=	`10 > 5`	`True`
Logical AND	`and`	`(5 > 2) and (3 == 4)`	`False`
Logical OR	`or`	`(5 > 2) or (3 == 4)`	`True`
Logical NOT	`not`	`not(10 == 10)`	`False`

TERNARY OPERATOR (SHORT-HAND)

Python provides a concise syntax for assigning values based on a single condition in one line:

```
age = 18
status = "Adult" if age >= 18 else "Minor"
```